UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the May/June 2009 question paper

for the guidance of teachers

4024 MATHEMATICS

4024/02 Paper 2, maximum raw mark 100

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UNIVERSITY of CAMBRIDGE
International Examinations

1	(a)		$\frac{4a^2+9}{6a}$ final answer	B1	1	
	(b)		$5b(b-2)$ final answer	B1	1	Condone missing final bracket After 0 + 0 give sc1 for <u>both</u> correct forms seen.
	(c)	(i)	(6, 2)	B1	1	Condone missing brackets
		(ii)	$\sqrt{(\pm 4)^2+(\pm 10)^2}$	M1		
			10.7 to 10.8	A1	2	Accept $2\sqrt{29}$
	(d)		For numerical $\frac{p \pm (or + or-)\sqrt{q}}{r}$			Completing the square
			$p = -11$ and $r = 6$ (or 2×3)	B1		B1 for $\frac{-11}{6}$; B1 for $\sqrt{\frac{205}{36}}$ oe
			$q = 205$ or $\sqrt{q} = 14.3$ to 14.32	B1		dep. on correct formula s.o.i or <u>used</u>
			0.55	B1		----------
			–4.22	B1	4	If final B0 + B0 then sc1 for 0.5 to 0.6 AND –4.2 to –4.22; or for any two answers given to 2 d.p.
					[9]	
2	(a)	(i)	Figs 378/the product of at least 2 of 20, 24, 7 and 60	M1		
			\$31.25	A1	2	Accept \$31.2 → \$31.3 \$0.000 031 2 → \$0.000 031 3 million;
		(ii)	$\frac{945-378}{378} \times 100$ or $\frac{945}{378} \times 100$	M1		Accept $\frac{78.125-31.25}{31.25} \times 100$
			150% cao	A1	2	
		(iii)	2:5 or m = 2, n = 5	B2	2	sc1 for partial simplification seen. 126:315, 54:135, 42:105, 18:45, 14:35, 6:15; or for $\frac{2}{5}$, or 1:2.5, or 5:2 or 2m:5m
	(b)		$\frac{480}{0.6} \times \left(\frac{2}{100} = 16\right)$	M1		
			16 cao	A1	2	sc1 for 9.6(euros) or (\$)800 seen.
					[8]	

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3	**(a)**	**(i)**	$\tan x = \frac{11}{4}$	**M1**		For any <u>complete</u> methods allow appropriate M and A marks. sc1 for 19.9 to 20
			70 to 70.02	**A1**	2	
		(ii) (a)	$\sin 28 = \frac{4}{PX}$ or $\frac{PX}{(\sin 90)} = \frac{4}{\sin 28}$	**M1**		GRADIAN ANSWERS **(i)** 77.80 sc1 for 22.2 or 12.2 **(ii)** **(a)** 9.39 … **(b)** 8.50 (leading to 2.5) or 7.77 … from Sine Rule (leading to 3.23)
			8.5 to 8.525	**A1**	2	
		(b)	$d = \frac{4}{\tan 28}$, or $PX \cos 28$, or $\frac{4 \sin 62}{\sin 28}$ or $\sqrt{PX^2 - 4^2}$	**M1**		
			7.5 to 7.6	**A1**		
			$11 - d$ (= 3.4 to 3.5)	**M1**	3	
	(b)		$r^3 = \frac{96}{\frac{4}{3}\pi}$ or 22.9 …	**M1**		
			2.84 to 2.841	**A1**	2 **[9]**	
4	**(a)**	**(i) (a)**	3 (lines of symmetry)	**B1**	1	
		(b)	order 3	**B1**	1	
		(ii) (a)	Use of $(9 - 2) \times 180$ etc.	**M1**	1	AG. Allow if 140° calculated, but not if quoted.
		(b)	$6x + 3y = 1260$ oe	**B1**		
			$y = 420 - 2x$ oe isw	**B1**	2	The second B mark implies the first.
		(c)	Sensible attempt at solving for x or y	**M1**		
			$x = 136$	**A1**	2	
	(b)	**(i)**	$\angle FEB = 114°$	**B1**	1	
		(ii)	$\angle BEA = 42°$	**B1**	1	
		(iii)	$\angle AGD = 63°$	**B1**	1 **[10]**	
5	**(a)**	**(i) (a)**	$\frac{1}{50}$, 0.02 cao	**B1**	1	Accept negatives
		(b)	$\frac{1}{2} \times (8 + 4) \times 200$ oe	**M1**		
			1200 m	**A1**	2	
		(c)	5 m/s	**B1**	1	
		(ii)	$150u = \frac{1}{2} \times 13 \times 150$ (= 975) oe	**M1**		
			$u = 6\frac{1}{2}$	**A1**	2	$\sqrt{4 + \frac{1}{2}}$ (their 5)

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	(b)	**(i)**	195 m	**B1**	1	
		(ii)	24.5 or (25.4 to 25.5) seen	**B1**		
			$\frac{\text{Distance}}{\text{Time}}$	**M1**		
			7.64 to 7.65	**A1**	3	**N.B.** $\frac{190}{25}$ = 7.6 scores the M1 only
					[10]	
6	**(a)**		$p = 11$ $q = 30$ $r = 60$ $s = 6$ all four	**B2**	2	sc1 for 2 or 3 correct
	(b)		$x = 2n + 1$ oe	**B1**		In **(b)**, accept any unsimplified form but –1, once, if not given explicitly
			$y = n(n + 1)$ oe	**B1**		
			$z = 2n(n + 1)$ oe √ $2 \times y$	**B1**	3	
	(c)		102	**B1**	1	
					[6]	
7	**(a)**	**(i)**	$\frac{2}{5}$ oe **fraction**	**B1**	1	Not 40%; 0.4
		(ii) (a)	$h = 25$	**B1**	1	
		(b)	$2(50 \times 15 + 60 \times 15) + 50 \times 60$	**M1**		
			6300 cm^2	**A1**	2	sc1 for 3300 or for 9300 cm^2
	(b)	**(i)**	$\frac{220}{360} \times 2\pi \times 9 \times 35$	**M1**		
			1208 to 1210	**A1**	2	
		(ii)	$\frac{220}{360} \times \pi \times 9^2$ (= 155.50 …)	**M1**		POSSIBLE GRAD ANSWERS **(ii)** 188 to 188.3 from $\frac{1}{2} \times 9^2 \times \sin 140$ (= 32.7 …); 177 to 178 from 81 × sin70 × sin20 (= 22.3 …) **(iii)** 4.9 from cos70; 6.2 from sin20 sc1 for 4.08 … or for 2.7 … --- If A0, then sc1 for 155 to 156 seen or for 25.9 to 26.1 seen
			$\frac{1}{2} \times 9^2 \times \sin 140$ (= 26.03 …)	**M1**		
			181 to 182	**A2**	4	
		(iii)	d = 9 – 9cos 70	**M1**		}
			= 5.92 to 5.93	**A1**	2	}sc1 for 3.07 to 3.08 seen
					[12]	

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8	**(a)**	**(i)**	$PQ = (x + 2)$ m			If **AB** used instead of x, –1 once
		(ii)	$BC = \frac{168}{x}$			
		(iii)	$QR = \frac{168}{x} + 11$ $\surd BC + 11$ all 3 (condone 10 + 1 for 11)	**B2**	2	sc1 for 1 or 2 correct
	(b)		Area = $(x + 2)\left(\frac{168}{x} + 11\right) - 168$ or $\surd PQ \times QR - 168$ as an expression in x	**M1**		or $(x + 2) + 10(x + 2) + 2 \times \frac{168}{x}$ oe
			correct working to $22 + 11x + \frac{336}{x}$	**A1**	2	Answer given
	(c)		$p = 158$ to $158\ \frac{1}{3}$	**B1**	1	
	(d)		Correct scales	**S1**		Condone reversed axes, if labelled
			7 correct plots (ignore $x = 9$) within 1 mm	**P1**		Accept if curve goes through correct points
			Smooth curve	**C1**	3	Not grossly thick; no straight lines Ignore curve for $x < 3$ and $x > 8$
	(e)		Clear attempt to draw tangent at (4, 150) gradient = –6 to –12	**T1** **G1**	2	Accept "integer" fractions
	(f)	**(i)**	143 ⩽ answer < 144	**B1**	1	
		(ii)	7.4 to 7.6	**B1**	1 **[12]**	
9	**(a)**	**(i)**	$\frac{AD}{\sin 38} = \frac{17}{\sin 114}$	**M1**		
			$AD = 17 \times \frac{\sin 38}{\sin 114}$	**M1 dep.**		----------
			11.4 to 11.5	**A1**	3	GRADIAN ANSWERS **(i)** 9.7 to 9.8 **(ii)** 140.9 to 141
		(ii)	$17^2 = 9^2 + 10^2 \pm (2) \times 9 \times 10 \cos x$ } or $\cos x = \pm [(9^2 + 10^2 - 17^2)/(2) \times 9 \times 10]$}	**M1**		
			$\cos C = \frac{10^2 + 9^2 - 17^2}{2 \times 9 \times 10} = (-0.6)$	**A1**		
			126 to 127	**A1**	3	

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	(b)	**(i) (a)**	$\overrightarrow{OQ} = \mathbf{p} + \mathbf{q}$	**B1**	1	In **(b) (i)**, –1, once, for unsimplified answers
		(b)	$\overrightarrow{RS} = \mathbf{p} - \mathbf{q}$ √ **(a)** $- 2\mathbf{q}$	**B1**	1	
		(c)	$\overrightarrow{OS} = 2\mathbf{p} + 2\mathbf{q}$ √ $2 \times$ **(a)**	**B1**	1	
		(d)	$\overrightarrow{OT} = 4\mathbf{p}$	**B1**	1	
		(ii)	(O, P and T are) collinear oe	**B1**		Marks in **(ii)** are dep on a correct **(i) (d)**
			$OT = 4OP$ oe	**B1**	2	
					[12]	
10	**(a)**		Correct scales and axes	**S1**		Condone reversed axes if clearly labelled
			Correct bases (width + position)	**B1**		
			Heights (2), 10, 8, 7, 4, 2	**H1**	3	
	(b)		$7 < t \leqslant 9$	**B1**	1	
	(c)		$(4 \times 2) + (10 \times 3.5) + (8 \times 4.5) +$ $(14 \times 6) + (8 \times 8) + (6 \times 10.5)$ (= 290)	**M1**		8, 35, 36, 84, 64, 63 Condone up to 3 slips
			$\div 50$	**M1**		Indep of first M
			5.8	**A1**	3	
	(d)	**(i)**	0	**B1**	1	Condone $\frac{0}{50}$, none, nil
		(ii)	$\frac{14}{25}$ oe 0.56	**B1**	1	
	(e)	**(i)**	$\frac{54}{175}$ $(= \frac{14}{25} \times \frac{27}{49})$ (0.308 to 0.309)	**B1**	1	In **(e)**, –1, once, for any answer not in lowest terms, or in decimal form
		(ii)	$\frac{88}{175}$ $(= 2 \times \frac{14}{25} \times \frac{22}{49})$ (0.502 to 0.503)	**B2**	2	sc1 for $\frac{44}{175}$ (0.251 to 0.252)
					[12]	
11	**(a)**	**(i)**	$\begin{pmatrix} -3 & 9 \\ -3 & 2x \end{pmatrix}$	**B2**	2	sc1 for 3 correct elements
		(ii)	$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ -1 + \frac{x}{3} & 1 \end{pmatrix}$ or $\mathbf{BA} = \begin{pmatrix} 1 & 3 - x \\ 0 & 1 \end{pmatrix}$			
			or $\mathbf{B}^{-1} = \begin{pmatrix} 0 & 3 \\ -1 & 3 \end{pmatrix}$ oe	**B1**		e.g. $3\begin{pmatrix} 0 & 1 \\ -\frac{1}{3} & 1 \end{pmatrix}$, (0.33 or better)
			$x = 3$	**B1 dep.**	2	

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	(b)	**(i) (a)**	SF = –2	**B1**	1	
		(b)	Centre is (1, 2)	**B2**	2	B1 for each coord. sc1 for $\begin{pmatrix}1\\2\end{pmatrix}$
		(ii)	Shear, x-axis inv., SF = 2	**B1** **B1**	2	Mention of a 2nd transformation loses both marks
		(iii) (a)	$\begin{pmatrix}1 & 2\\0 & 1\end{pmatrix}\begin{pmatrix}k\\2\end{pmatrix}=\begin{pmatrix}k+4\\2\end{pmatrix}$ $k = 4$	**MA1**	1	
		(b)	ET(L) = E((8, 2)) = (–13, 2)	**B2**	2 **[12]**	sc1 for (–2.5, 2)

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